

YASHWANTH REDDY MALI

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SUMMARY

Backend engineer, 2 years shipping production systems in Go, Python, Java, and TypeScript. Built a runtime eBPF firewall that catches AI agents exfiltrating data, a Zanzibar-style auth engine (sub-10ms P95 at 10K RPS), a Rust SQL engine (3.2x on TPC-H), and the full backend of a production AI data-discovery platform. Contributor to Auth0 OpenFGA and Google Malloy.

EXPERIENCE

Software Engineer - Credible Data, Boulder, CO

Aug 2025 - May 2026

- Shipped the Express 5/Node.js backend for Credible Atlas, an AI-native data-discovery platform: Server-Sent Events chat streaming, dataset search, and Malloy execution over PostgreSQL 16, Neo4j graph lineage, and BigQuery.
- Cut API P95 **60%, 1.2s to 480ms** (production): connection-pool tuning, EXPLAIN ANALYZE on hot query plans, and index redesign that eliminated sequential scans on high-cardinality columns.
- Guaranteed at-least-once delivery with idempotent consumers via a graph outbox pattern spanning PostgreSQL and Neo4j - **zero data loss**, full audit lineage on every state mutation.
- Routed natural-language questions through the Claude Agent SDK and a custom MCP server over JSON-RPC; secured with an Auth0/JWT gateway and role-scoped middleware across **10+ route groups**.
- Deployed GCP infrastructure (Compute Engine, Cloud Run) with GitHub Actions CI/CD, Secret Manager, and health-check-triggered rollback for zero-downtime deploys.

Software Engineer Intern - IntelWings, Chandigarh, India

Feb 2023 - Jun 2024

- Shipped a transaction-monitoring backend (Java/Spring Boot) processing **100K+ monthly transactions** at **sub-200ms** (production); owned reliability: parameterized SQL, PII masking, audit logging.
- Cut query latency **70%, 800ms to 240ms** (production), with multi-tier caching (Redis + local).
- Re-architected deployment pipelines (Jenkins, Docker, Kubernetes); shrank release cycles from **5 days to 3**.

PROJECTS

Dispatch: Multi-Tenant Webhook Delivery Platform | Go, PostgreSQL, Redis, Kafka | GitHub May 2026 - Jun 2026

- Fixed a probe-stampede race in the half-open circuit breaker: a conditional UPDATE single-flights one real signed event to a recovering endpoint.
- Ran retries through a single Kafka retry topic with retry_after; exhausted events land in a Postgres DLQ - chosen over a Kafka DLQ topic so replay gets filtering and pagination.
- Load-tested with vegeta: **100% 202s** at 50 events/s, ingest **p99 2.1 ms**, delivery **p99 ~2.5 ms** (local Docker Compose, not cloud SLOs).
- Instrumented with Prometheus, Grafana, and OpenTelemetry traces to Jaeger; kept tenant_id off ingest metric labels to protect cardinality.

Interlock: Runtime AI-Agent Exfiltration Firewall | Go, eBPF, Kubernetes | GitHub

Feb 2026 - Present

- Wrote a Go MCP proxy that tracks secrets through base64/hex/URL/gzip encodings and refuses to forward exfiltrating tool calls.
- Traced connect()/write() syscalls with eBPF (cilium-ebpf) to catch side channels the proxy can't see; containment defers ~100 ms so payload proof can land, then SIGKILL.
- Deployed as a sensor-only Kubernetes DaemonSet, validated on EKS; exposes Prometheus metrics, Slack/PagerDuty trip webhooks, OCSF SIEM export, SIGHUP hot reload.
- Backed every detector with known-gap tests naming what it does not catch - **178 tests**, race-enabled CI; overhead measured as engine-on vs. passthrough delta, **~0.5 ms** on sensitive reads (local benchmark).

Zenith: Distributed Authorization Engine | Go, CockroachDB, gRPC, Redis | GitHub

Sep 2025 - Feb 2026

- Wrote a Zanzibar-inspired authorization engine with recursive permission expansion across **10K+ tenants**.
- Hit **sub-10ms P95 at 10,000+ req/sec** (benchmark, k6) with singleflight deduplication and LRU caching.

LineageGraph: GraphRAG Data Lineage | Python, LangGraph, FastAPI, PostgreSQL | GitHub Feb 2025 - Jul 2025

- Wired a GraphRAG pipeline: local LLM inference (Ollama) with vector embeddings, verified against PostgreSQL recursive CTEs to block hallucination; OpenTelemetry tracing across 4 agent nodes, **sub-200ms queries** (local).

OPEN SOURCE

- Landed PRs in **Auth0 OpenFGA** and **Google Malloy**: bug fixes across query planning and validation, including a consistency-token correctness fix in OpenFGA's Check API.

TECHNICAL SKILLS

Languages: Go, Python, Java, TypeScript, Rust, SQL, C (eBPF)

Backend & APIs: Spring Boot, FastAPI, Express, Node.js, gRPC, REST, MCP, LangGraph

Infrastructure & Cloud: AWS (EKS), GCP (Compute Engine, Cloud Run), Docker, Kubernetes, Git, GitHub Actions, Jenkins

Data & Messaging: PostgreSQL, Kafka, Redis, BigQuery, Neo4j, CockroachDB

Observability & Testing: OpenTelemetry, Prometheus, Grafana, Jaeger, k6, vegeta, race-enabled CI

Security: eBPF, OAuth2, JWT, HMAC signing, OCSF/SIEM export

EDUCATION

University of Colorado Boulder | M.S. Computer Science, GPA 3.9/4.0

Aug 2024 - May 2026

Osmania University | B.E. AI & Data Science, GPA 8.7/10.0

Aug 2020 - May 2024